Iterative Nonlinear Computation Underlying Abstract Reasoning

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iQuest Research

X-AGI
X-AGI 2026 Poster

1. Motivation / Problem

- Universal Transformers (UTs) are strong on abstract reasoning tasks such as ARC-AGI, Sudoku, and Maze, but the source of their gains is unclear.
- This work asks which structural properties matter most.
- Our main finding: the gains come primarily from recurrent parameter sharing across depth and strong nonlinearity inside the Transformer block, rather than from elaborate architectural heuristics.

2. Core Idea / Contributions

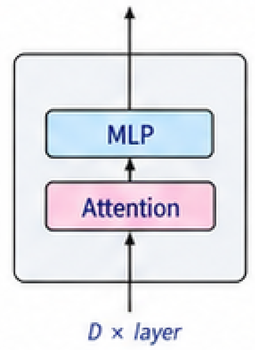
Insight
Abstract reasoning benefits from iterative nonlinear computation.

ConvSwiGLU
Augment the SwiGLU feed-forward block with a channel-wise short convolution plus post-convolution SiLU to strengthen nonlinearity and local token interactions.

TBPTL
Truncated Backpropagation Through Loops trains long recurrent rollouts more stably by freezing gradients in early loops and backpropagating only through later loops.

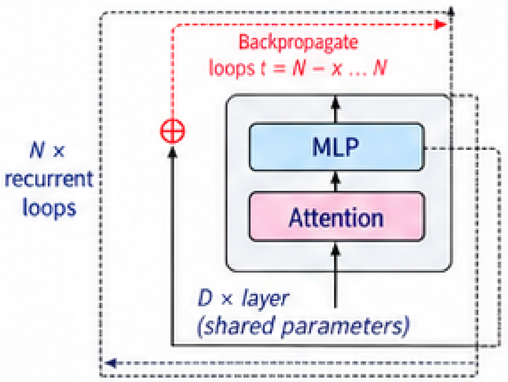
3. URM Architecture

Standard Transformer Block



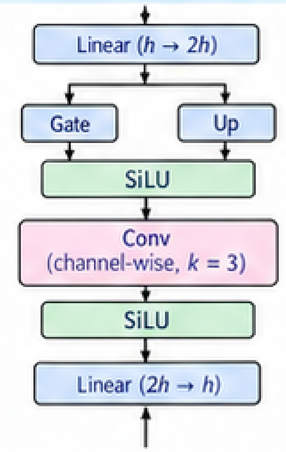
Standard Transformer (one-pass)

Universal Reasoning Model (URM)



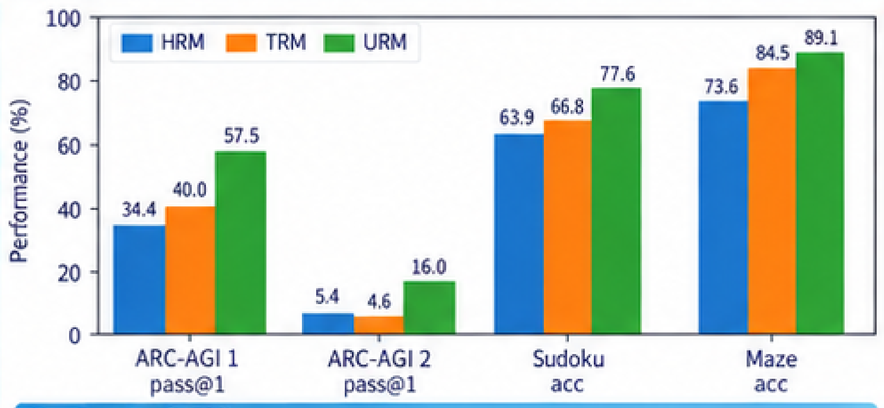
Forward only
loops t = 1 … N − x

ConvSwiGLU Module



URM strengthens iterative reasoning by combining recurrence, stronger nonlinearity, and stable loop training.

4. Main Results on Abstract Reasoning



Key Results (URM)

	ARC-AGI 1			ARC-AGI 2			Sudoku	Maze
	pass@1	pass@10	pass@100	pass@1	pass@10	pass@100	acc	acc
URM	57.5	74.4	82.4	16.0	26.9	34.3	77.6	89.1

Key Performance (URM)

57.5% ARC-AGI 1 pass@1

16.0% ARC-AGI 2 pass@1

77.6% Sudoku accuracy

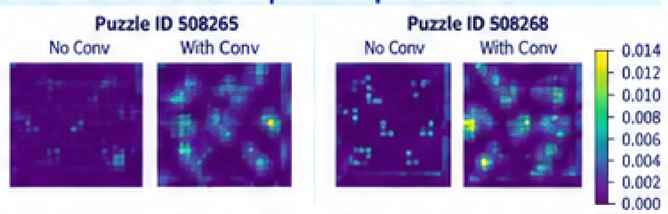
89.1% Maze accuracy

5. Ablation / Why It Works

A. Component Ablation (ARC-AGI 1)

Model	pass@1	pass@10	pass@100
Full URM	**57.5**	**74.4**	**82.4**
- SiLU after Conv	50.2	64.5	75.0
- ConvSwiGLU	45.3	62.6	72.0
- TBPTL	40.0	54.4	64.5

B. Short Convolution Improves Expressiveness



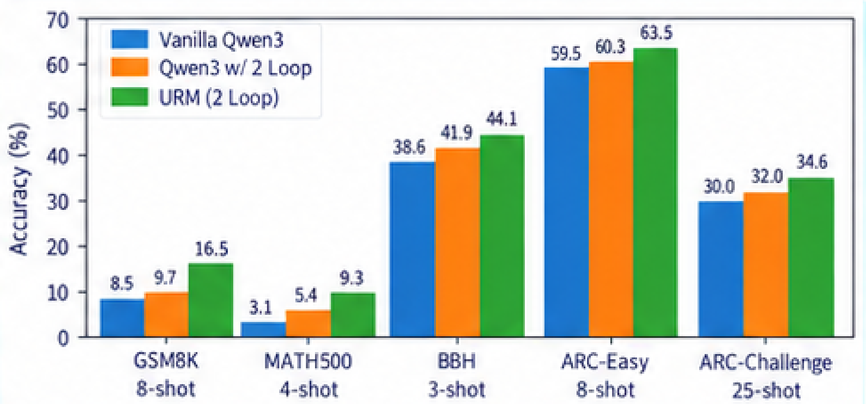
Adding short convolution yields more diverse and structured attention patterns, indicating stronger channel mixing.

C. TBPTL Helps Stable Long-Horizon Learning

Total Loops	Truncate First k	pass@1	pass@10	pass@100
8	1 (best)	50.00	68.63	78.01
16	4 (best)	57.50	76.75	84.50

Takeaway: moderate truncation improves optimization stability without destroying long-horizon refinement.

6. Transfer to Qwen3-8B (300B pretraining tokens)



The same design principle transfers beyond grid tasks, consistently improving downstream reasoning in a matched 8B LLM.

7. Take-Home Message / Conclusion

1. Strong abstract reasoning does not mainly come from larger models or ad hoc heuristics; it comes from iterative nonlinear computation.
2. URM combines shared recurrent refinement, ConvSwiGLU, and TBPTL to substantially improve reasoning benchmarks.
3. The same principle also improves a pretrained 8B language model across multiple reasoning tasks.